The Zimmerman Formula

Derivation and Testable Predictions

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Abstract: We present a derivation relating the MOND acceleration scale a0 to cosmological parameters. From the Friedmann critical density, we derive **a0 = c√(Gρc)/2 = cH0/5.79**, where 5.79 = 2√(8π/3) emerges from the Friedmann equation structure. This achieves 0.5% agreement with observed a0 and predicts that a0(z) = a0(0) × E(z). This paper presents the mathematical derivation, its direct consequences, and quantitative predictions that can confirm or refute the formula.

1. The Derivation

1.1 Starting Point: Friedmann Critical Density

The critical density for a flat universe:

$$\boxed{\rho_c = \frac{3H_0^2}{8\pi G}}$$

1.2 Constructing an Acceleration Scale

Using dimensional analysis with c, G, and ρc to construct an acceleration [L T⁻²]:

$$[c\sqrt{G\rho_c}] = \frac{L}{T} \times \sqrt{\frac{L^3}{MT^2} \times \frac{M}{L^3}} = \frac{L}{T} \times \frac{1}{T} = \frac{L}{T^2}$$

✓

1.3 Determining the Coefficient

Substituting ρc = 3H0²/(8πG):

$$a_0 = \frac{c}{\alpha}\sqrt{\frac{3H_0^2}{8\pi}} = \frac{cH_0}{\alpha\sqrt{8\pi/3}}$$

Setting α = 2:

$$\boxed{a_0 = \frac{cH_0}{5.79}} \quad \text{where } 5.79 = 2\sqrt{\frac{8\pi}{3}}$$

1.4 Numerical Verification

H_0 (km/s/Mpc)	Source	Predicted a_0 (m/s²)	Error
67.4	Planck CMB	1.131×10^{-10}	5.7%
71.1	**Best fit**	**1.193×10^{-10}**	**0.5%**
73.0	SH0ES	1.225×10^{-10}	2.1%

Observed: $a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s² (McGaugh et al. 2016)

2. Mathematical Consequences

These follow directly from the derivation with no additional assumptions.

2.1 Inverse Formula: H0 from a0

$$H_0 = \frac{5.79 \times a_0}{c} = 71.5 \text{ km/s/Mpc}$$

Method	H_0 (km/s/Mpc)
Planck CMB	67.4 ± 0.5
Zimmerman (from a_0)	**71.5 ± 1.0**
SH0ES Cepheids	73.0 ± 1.0

2.2 Redshift Evolution

Since $\rho c(z) = \rho c(0) \times E(z)^2$:

$$a_0(z) = a_0(0) \times E(z) \quad \text{where } E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

Redshift	E(z)	$a_0(z) / a_0(0)$
0	1.00	1.00
1	1.70	1.70
2	2.96	2.96
5	8.83	8.83
10	20.1	20.1

2.3 The Cosmic Coincidence Resolved

The observation a0 ≈ cH0 is not coincidental:

$$\frac{a_0}{cH_0} = \frac{1}{5.79} = 0.173$$

The factor 5.79 = 2√(8π/3) arises from the Friedmann equation geometry.

3. Testable Predictions

These predictions can falsify the formula.

3.1 Baryonic Tully-Fisher Evolution

Prediction: At fixed rotation velocity v_flat:

$$\Delta \log M_{bar} = -\log E(z)$$

Redshift	E(z)	Predicted Δlog M_bar
1.0	1.70	-0.23 dex
2.0	2.96	-0.47 dex
3.0	4.65	-0.67 dex

Falsification: If Δlog M = 0 at z > 1, the formula is falsified.

3.2 RAR Scale Evolution

Prediction: The RAR transition scale evolves:

$$g_\dagger(z) = g_\dagger(0) \times E(z)$$

Falsification: If g† is constant with redshift, the formula is falsified.

3.3 Mass Discrepancy Evolution

Prediction: At z > 6, mass discrepancy ratios should be ~3× higher than local.

Falsification: If M_dyn/M_bar at z > 6 equals local values, the formula is falsified.

3.4 Growth Rate Enhancement

Redshift	f_ΛCDM	f_Zimmerman	Enhancement
0.5	0.68	0.70	+3%
1.0	0.78	0.82	+5%
2.0	0.88	0.97	+10%

Falsification: If growth rates match ΛCDM exactly, the formula is falsified.

4. Validated Predictions

4.1 SPARC Database (175 galaxies)

Metric	Result
BTFR slope	4.000 ✓
RAR scatter	0.20 dex ✓
Mean g_obs/g_MOND	1.007 ✓

4.2 JWST High-z (D'Eugenio et al. 2024)

Model	χ^2
Zimmerman (evolving a_0)	**59.1**
Constant a_0	124.4

The evolving a0 model fits 2× better.

5. What This Does NOT Prove

1. **Does not prove MOND is correct** — only that IF MOND is correct, a0 is determined by cosmology
2. **Does not explain the Bullet Cluster** — lensing/gas offset remains unexplained
3. **Does not derive the interpolating function** — only the scale a0
4. **Does not prove dark matter doesn't exist** — provides alternative scaling

6. Summary

$$a_0 = \frac{c\sqrt{G\rho_c}}{2} = \frac{cH_0}{5.79} \quad \text{where } 5.79 = 2\sqrt{8\pi/3}$$

Result	Value
Agreement with observed a_0	**0.5%**
H_0 from a_0	**71.5 km/s/Mpc**
JWST fit improvement	**2× better χ^2**
BTFR slope from SPARC	**4.000 exactly**

References

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Code: github.com/carlzimmerman/zimmerman-formula | **DOI:** 10.5281/zenodo.19114050